Log returns

In the financial literature, log returns are defined as $\ln(p_i) - \ln(p_j)$.

It is commonly that stocks returns tend to be normally distributed (with fat tails as Fama and French proved). If returns are small enough log returns are well approximated by simple returns, and therefore, log returns can be also approximated with a Normal distribution and prices are log-normal, since it's logarithm follows a normal distribution. Models such as Black-Scholes make the assumption of log-normal prices.

Being R the simple return, and using Taylor expansion:

$R_i = \frac{p_i - p_j}{p_j}$ (normally distributed)

$R = e^{\ln(1+R)} - 1 = \ln(1+R) + \frac{\ln(1+R)^2}{2} + \cdots \approx \ln(1+R)$ (if small, we can use log returns)

By definition:

$1 + R_i = p_i / p_j$

Therefore:

$R \approx \ln(1+R) = \ln\left(\frac{p_i}{p_j}\right) = \ln(p_i) - \ln(p_j)$ (log returns normally distributed, prices log-normal)

Instead of starting with the assumption of normal returns, you could start with the assumption of log-normal prices and under the same conditions prove the inverse.

Remember that this is only a good approximation if the period in which you analyze the returns has small enough returns. For example using this for the daily returns of cryptocurrencies would be useless.

Drift and volatility

Drift and volatility stand for the mean and standard deviation of the returns. If anything is specified, these drift and volatility are interpreted as the annualized ones.

There are other types of volatility estimators that take in account a wider variety of factors. For example the volatility estimator of Parkinson or Garman-Klass, that take into account the intraday information from OHLC charts.

$$\hat{\sigma}^2_{Parkinson} = \frac{1}{4Tln(2)} \sum (\ln(H_t/L_t))^2$$

$$\hat{\sigma}_{GK}^2 = \frac{1}{T}\sum[0.5(\ln(H_t/L_t))^2 - (2ln2 - 1)(\ln(C_t/O_t))^2]$$

T: number of days in the sample period

H_t: Highest price of the period

L_t: Lowest price of the period

C_t: Close price of the period

O_t: Open price of the period

<u>Introduction to stochastic processes</u>

Stochastic processes are time-dependent random variables which can be continuous or discrete. Time series are samples at uniform intervals (discrete) of a stochastic process. A random walk is the simplest time-series model.

<u>Simple random walk</u>

In a simple random walk each increment is a random IID variable z_i (for example assume it as a standard random variable: mean 0 and variance 1), and it doesn't depend on other increments in the past. Examples of random variables following the z_i distribution example could be a fair toss coin (+1 or +1 with equal probability) or a Gaussian random variable $z_t \sim N(0,1)$.

S_i represents the simple random walk, which follows:

$$S_0 = 0$$

$$S_t = z_1 + z_2 + \cdots + z_t$$

S_i has the Markov property because it only depends on S_{i-1}. ($S_i = S_{i-1} + z_i$)

$E[S_t] = 0$

$Var[S_t] = t$

Since the variance is not constant this process is non-stationary, but as we have seen before differentiating/computing increases we could make it stationary.

<u>Generalized random walk</u>

To generalize the previous random walk, we can modify the increase to have a certain mean μ and variance σ^2 (also the covariance between different periods is 0): $r_t = \sigma z_t + \mu$.

Let X_i denote a generalized random walk which follows the same initial conditions than S_i:

$E[X_t] = t\mu$

$Var[S_t] = t\sigma^2$

Characteristics for identifying a random walk in your data: lack of serial correlation(ACF).

Monte Carlo simulation for random walks

Monte Carlo methods are used to numerically compute mathematical expressions difficult to evaluate. In the case of a random walk the expected value is very easy to compute but in more complex models it could be very difficult to compute it deterministically. For example, knowing the drift and volatility we can generate a huge number of simulations and then study the properties of the distribution of our simulated terminal prices.

The sampling error will decrease with the sample size

Time series models

Time series models try to model variables with more complex models than a random walk. Among them stand out models such ARIMA or GARCH, but you can find many more or even create your own ones.

If we model a variable with AR, MA, or ARIMA models, it depends on past variables and it is not IID anymore. By construction they are stationary models (all the moments of the distribution are constant) and all of them have a well-defined mean, variance and autocorrelation function.

Although these processes are stationary, in practice it is a too strong condition. It is more feasible to just check weakly stationarity, in which the first and second moments are constant <--> constant mean and variance, so from now on I will refer to weak stationarity simply as stationarity.

More on, in stationary series, the covariance with its lagged variables depends only on the difference of time, not its position. For example, $Cov(Y_1, Y_9) = Cov(Y_4, Y_{12})$.

Autoregressive model (AR):

$$AR(p): R_t = c0 + c1R_{t-1} + \cdots + c_pR_{t-p} + \sigma z_t \qquad z_t \sim IID(0,1)$$

Autoregressive process of order p "AR(p)" defines a variable which depends on past values of itself and some randomness. This random variable (z_t) is usually defined as a standardized one coming from an IID distribution. We want to compute the coefficients c that minimize the mean squared

error. AR is just like a linear regression, since we are minimizing the same objective function and therefore we can use the same procedures and well-defined results: OLS, maximum likelihood or Yule-Walker solutions, which are identical.

Recall that when you use linear regression there exist some methods to avoid overfitting such as:

1. Train with more data.

2. Cross-validation: Divide the data into k subsets called folds, and iteratively train the algorithm on k-1 folds while using the remaining fold as the test set.

3. Regularization: Change objective function to Lasso, Ridge or Elastic Net.

For other types of data you could also use data augmentation and feature selection.

<u>An special case:</u>

To understand how this series work, let's solve for a special case: AR(1)

$$R_t = c0 + c1R_{t-1} + \sigma z_t$$

$$E[R_t] = c0 + c1E[R_{t-1}] = (using\ stationarity) = \frac{c0}{1 - c1}$$

For convenience we can define two new parameters and rearrange the equation defining the AR(1) process:

$\mu = \frac{c0}{1-c1}$ $\quad \lambda = -c1$

AR(1): $R_t - \mu = -\lambda(R_{t-1} - \mu) + \sigma z_t, \qquad |\lambda| < 1$

We compute the change in the random variable relative to its mean. This model is often used as a mean reversion model.

$$Var[R_t] = \gamma_0 = E[(R_t - \mu)^2] = substituting\ with\ the\ definition\ of\ AR(1) = (-\lambda)^2 E[(R_{t-1} - \mu)^2] + \sigma^2 E[z_t^2]^* = \lambda^2 \gamma_0 + \sigma^2$$

$$Var[R_t] = \gamma_0 = \frac{\sigma^2}{1 - \lambda^2}$$

*Note that when we expand the square the cross terms drop out due to the independence of z_t

$$E[z_t(R_{t-k} - \mu)] = 0$$

The covariance function of the returns with other previous returns can also be defined:

$$Cov[R_t, R_{t-k}] = \gamma_k = E[(R_t - \mu)(R_{t-k} - \mu)] = -\lambda E[(R_{t-1} - \mu)(R_{t-k} - \mu)] = -\lambda \gamma_{k-1}$$

$$\gamma_k = (-\lambda)^k \gamma_0 = \frac{(-\lambda)^k}{1-\lambda^2}\sigma^2$$

Notice that as I mentioned before, the covariance depends on k (time difference) but not on time itself.

Due to stationarity the variance of different R_t is the same, so the correlation between two lagged returns can be expressed as:

$$p_k = Corr(r_t, r_{t-k}) = \frac{Cov(r_t, r_{t-k})}{\sqrt{Var(r_t)Var(r_{t-k})}} = \frac{\gamma_k}{\gamma_0}$$

This definition for correlation holds for any stationary model, where γ_k is the covariance for the defined model.

<u>Moving average process (MA):</u>

$$MA(p): R_t = \mu + \Phi_1 z_{t-1} + \cdots + \Phi_q z_{t-q} + \sigma z_t \quad , \quad z_t \sim IID(0,1)$$

In moving average processes of order q "MA(q)", returns (or the variable you want to model) are expressed as a linear combination of some random shocks $\Phi_q z_{t-q}$ plus the mean μ and some randomness σz_t. These random shocks can be decomposed in Φ, which are the coefficients to be computed, and z, representing the random process composing the model.

As you may have thought, the random shocks are not directly observable so we can't just fit a linear regression in the common way. To compute the parameters you might use some other methods in which I won't study here, such us Dublin's or maximum-likelihood method.

As a curiosity and because you can also transform MA processes into AR ones to compute it's parameters, it can be proved that a AR(1) process is equivalent to an MA(∞) process.

AR(1): $R_t - \mu = -\lambda(R_{t-1} - \mu) + \sigma z_t, \qquad |\lambda| < 1$

Let $Y_t = (R_t - \mu)/\sigma$, and substitute in the AR(1) equation:

$$Y_t = z_t - \lambda Y_{t-1} = z_t - \lambda[z_{t-1} - \lambda Y_{t-2}] =$$

$$= z_t - \lambda z_{t-1} + \lambda^2 Y_{t-2} = z_t - \lambda z_{t-1} + \lambda^2[z_{t-2} - \lambda Y_{t-3}] =$$

$$= z_t - \lambda z_{t-1} + \lambda^2 z_{t-2} - \lambda^3 Y_{t-3} = z_t - \lambda z_{t-1} + \lambda^2 z_{t-2} - \lambda^3[z_{t-3} - \lambda Y_{t-4}] = \cdots$$

Doing this indefinitely:

$$Y_t = \sum_{k=0}^{\infty} (-\lambda)^k z_{t-k}$$

Autoregressive moving average model (ARMA)

$$ARMA(p,q): R_t = co + c1R_{t-1} + \cdots + c_p R_{t-p} + \sigma z_t + \Phi_1 z_{t-1} + \cdots + \Phi_q z_{t-q}$$

$z_t \sim IID(0,1)$

ARMA(p,q) model is just a combination of the AR and MA models, and it is the model that tends to work better.

If our data series were not stationary we can use ARIMA, where I represent order of integration. This because a good way for transforming non-stationary data in stationary. Intuitively, when we take the increase(returns) instead of the values(stock prices), we can remove some of the trend or seasonality of the data. As I analyzed in one of the previous chapters, prices didn't tell anything interesting, but returns (first order integration) could be expressed with a very surprising normal distribution (to be exact, I should take into account the fat tails). The order of differentiation is an still argued topic, not for the models mentioned before, but for the loose of information: in some of the talks, Marcos Luiz mentioned that when we differentiate these series, we lose information in exchange of being able to model them, but always the research literature differentiated once as a rule of thumb. He suggested that taking fractional differences we could reach an optimal point in which we don't lose too much information. (I won't enter in this but is just a thing I found interesting for the idea itself, and to show how deeply have some topics gone that the research topics only in this field are almost infinite).

To check stationarity of the date we can use a Dicky-Fuller test, and using partial-autocorrelation (PACF) and autocorrelation (ACF) you can have an intuition for the parameter (p and q) that would fit better in the model, as I will explore I EL SIGUIENTE PUNTO.

Choosing a model and its order

We have seen a bunch of different proposals to model the stocks market, but which one is the best? We proved that random walks don't seem to be representative but how to pick a model that correctly models it?

Given a dataset of observed returns, we can compute the sample auto covariance (covariance with k previous values) similarly to the unbiased estimator for the variance:

$$\hat{\gamma}_k = \frac{1}{T-1}\sum_{t=k+1}^{k+T}(r_t - \hat{\mu})(r_{t-k} - \hat{\mu}')$$

Observing the serial correlation of the data will give an intuition of its behavior: compute the sample correlation between a return and a lagged return.

$$\widehat{p_k} = \frac{\widehat{\gamma_k}}{\widehat{\gamma_0}}$$

If it is different than 0 it means that there is some flow of information between periods and returns are not following a random walk.

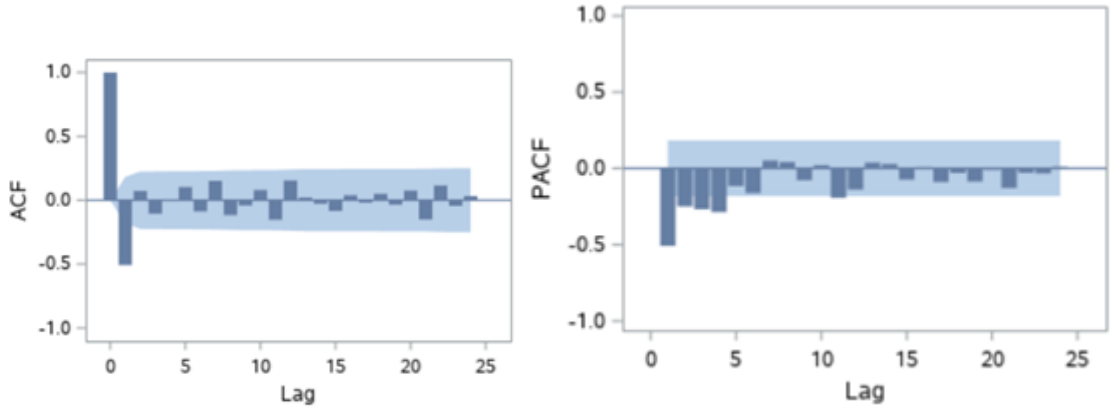
To select a model (and also its order) I will use a simple but very useful criterion using both PACF and ACF functions. On one hand, ACF is the partial autocorrelation function; basically we see the correlation between a variable and its previous values. On the other hand PACF is the partial autocorrelation function, in which we see a plot of the partial correlation. As you may know (bivariate) correlation is used to measure a linear relationship between two variables. The main problem we could have with this correlation is that we could get that two variables are correlated when in fact they are not, and there is a third variable that is giving a false sense of correlation. With partial autocorrelation we compute the correlation between two variables but controlling the others (like a conditional correlation) so we can eliminate the effect of a third variable.

The following table explains how to select the order based on these functions.

Terms	ACF	PACF
AR	Geometric	p significant lags
MA	q significant lags	Geometric
ARMA	Geometric	Geometric

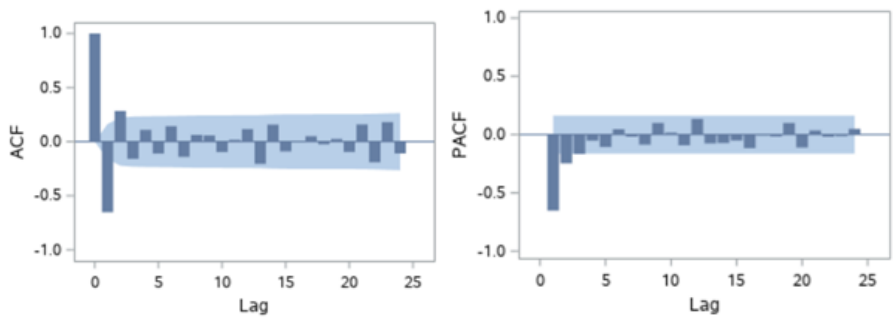
AR models tend to have a gradual decrease in the ACF and a sharp decrease after in the PACF after p significant lags. The opposite happens for MA. For the ARMA models both plots descend gradually.

Here there are some examples:



If the data was stationary (remember that it can be checked with a Dickey-Fuller test), this plots would suggest an MA(1) model.

For an ARMA model both plots would descend gradually and we should pick in a lag which seems to decrease more abruptly.



In this example we could try with an ARMA(1,1) or an MA(2).

But this is just an indicator, it is very common to find plots that aren't that clear so the best think I suggest is to try a few models that you see likely to fit and then evaluate them with some well-known criterions to compute the likelihood of a model such as the Akaike or the Bayesian criterion, which won't be explained here.

An ARIMA model is just an ARMA model but after differentiating to make the data stationary.

<u>Binomial tree</u>

In this model asset prices can move up or down and we need to compute the probability and returns of each direction. It is based on random walks, and is extremely simplistic.

At each step we have returns are defined as $r_t = \ln\left(\frac{S_t}{S_{t-1}}\right) = a + bz_t$ where $z_t = \pm 1$

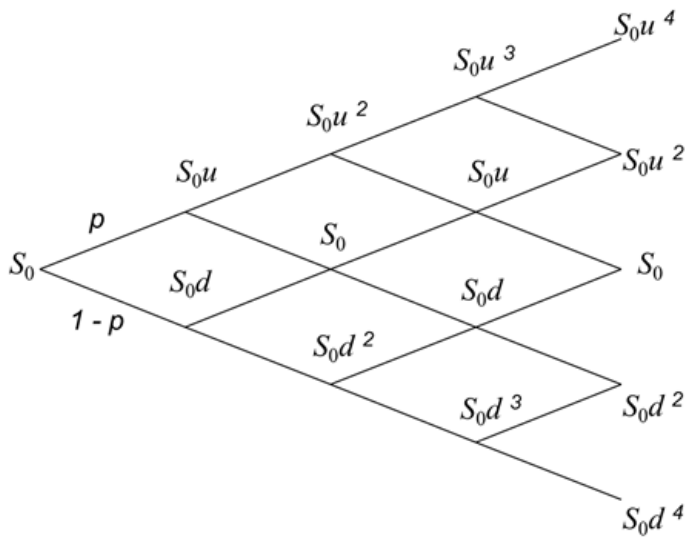
Similarly to random walks $\ln\left(\frac{S_t}{S_0}\right) = r_1 + r_2 + \cdots + r_T = aT + b\sum z_t$

This model can be used to price a certain type of options, as well as to compute expected values, default probabilities, risk, duration, and study boundaries.

Being S_0 the initial asset price, $S_{t+1} = \begin{cases} S_t * u \; with \; probability \; p \\ S_t * d \; with \; probability \; q = 1 - p \end{cases}$

u and d are price multipliers, and from a combinatorics view (identical to the binomial distribution), we can compute the probability of achieving a certain quantity at S_t.

$$Prob(S_t = S_0 u^k d^{t-k}) = \binom{t}{k} p^k (1-p)^{t-k}$$



<u>Gamblers ruin</u>

Assume you start with a certain amount of money x, and at each step you have two options: win some money (b) with probability p or lose the same amount (b) with probability $q = 1 - p$.

This problem answers the questions "What is the probability of losing everything ($x = 0$) before arriving to capital a ($x = a$)?" and "What is the expected duration of the game?"

Being Q_x the probability of ruining before getting the capital a, can be intuitively expressed in a recursive way:

$$Q_x = pQ_{x+b} + (1-p)Q_{x-b}\,,\;\; 0 < x < a$$

It is a difference equation and gives the following solution:

$$Q_x = \frac{\left(\frac{q}{p}\right)^{\frac{a}{b}} - \left(\frac{q}{p}\right)^{\frac{x}{b}}}{\left(\frac{q}{p}\right)^{\frac{a}{b}} - 1} if\ p \neq q$$

$$Q_x = 1 - \frac{x}{a}\ \ if\ p = q$$

With a=infinite

$$Q_x = \begin{cases} 1, & p \leq q \\ \left(\frac{q}{p}\right)^x, & p > q \end{cases}$$

To compute the expected duration of the game D_X, it can be derived in a similar way as before

$$D_x = pD_{x+1} + (1-p)D_{x-1}\ ,\ \ 0 < x < a$$

It has lots of applications in topics such as gambling, insurance, credit default or bet sizing.

<u>Stochastic calculus</u>

Until now we have been working with discrete time series. Our random walk is a stochastic process that evolves with time formed by a set of random steps.

Now we will see the basis of stochastic calculus applied to quantitative finance and derive some of its more important results such as the Black-Scholes-Merton equation.

<u>Brownian motion</u>

Take a random walk as our starting point. Between each discrete step a definite amount of time passed, but now I will make the time between these random steps smaller and smaller. If we make that each step happens in an infinitesimal time, we can generate a continuous random walk.

Remember the previous random variable z_t that was defined each time step. It had $E[z_t] = 0, Var(z_t) = 1, Cov(z_t, z_s) = 0$

Let a sum of a certain number of this random variable be defined as $B_{1,T} = \sum_{t=t_0+1}^{t_0+T} z_t$ where $(t_0 + 1)$ is the starting time and T is the number of steps made by the summation. The 1st sub index (1) refers to the increment at each time step, which is of one unit in this case.

$E_t[B_{1,T}] = 0,$

$Var_t(B_{1,T}) = T$

$Var_{t_1}(B_{1,T}) = T - (t_1 - t_0),\ t_0 \leq t_1 \leq T$

This last variance means the variance of the process $B_{1,T}$ computed at time t_1. Since it is calculated in the middle of the process, the information before t_1 has no relevance and we only take the time steps above it.

To make the process continuous it is necessary to take limits, but not in any way: mean and variance have to be finite.

<u>Standard Brownian motion: scaling the random walk</u>

I will present some ways in which we could take limits, rescaling some of its variables, and you will see that this limits can't be computed in any way.

Case 1: rescale time step only

Let $\Delta t = \frac{T}{n}$ go to 0. Δt is the time step, T the total time (not number of steps anymore) and n the number of steps.

$$B_{\Delta t,T} = \sum_{t=1}^{n} z_t$$

$$E\left[B_{\Delta t,T}\right] = 0$$

$$Var\left(B_{\Delta t,T}\right) = nVar(z_t) = n$$

$$\lim_{n\to\infty} n = \infty$$

Since T is fixed, when we take the limit of Δt the number of steps (n) becomes ∞ and we get a process with infinite variance.

Case 2: rescale time step and step size

Let $\Delta t = \frac{T}{n}$, $\varepsilon_t = \lambda z_t$

The term ε_t multiplies the size of the step by the factor λ.

$$B_{\Delta t,T} = \sum_{t=1}^{n} \varepsilon_t = \lambda \sum_{t=1}^{n} z_t$$

$$E\left[B_{\Delta t,T}\right] = 0$$

$Var(B_{\Delta t,T}) = nVar(\varepsilon_t) = n\lambda^2$

The mean of the process is the same as before, but the variance depends on λ. As $n \to \infty$ a right election of λ can make the variance finite. Suppose $\lambda = \frac{1}{n}$

$\lim_{n\to\infty} Var(B_{\Delta t,T}) = 0$

This is not a good election for λ because the variance would converge to 0 and we would not have a random variable anymore.

Case 3: rescale time step and step size simultaneously in a specific relationship:

Let $\Delta t = T/n, \ \ \lambda = \sqrt{\Delta t}, \ \ \varepsilon_t = \lambda z_t$

$$B_{\Delta t,T} = \sqrt{\Delta t}\sum_{t=1}^{n} z_t$$

$E\left[B_{\Delta t,T}\right] = 0$

$Var(B_{\Delta t,T}) \ = T$

Choosing the right value for λ both mean and variance are finite.

Intuitively $B_{\Delta t,T}$ is the sum of random variables (with finite mean and variance) and therefore applying CLT it follows a Normal distribution for a sufficient sample size.

$$\lim_{n\to\infty} B_{\Delta t,T} \sim N(0,T)$$

This new motion is called standard Brownian motion/Wiener process, and its main properties are that it is everywhere continuous, nowhere differentiable (fractal paths), and it converges in the distribution.

Also it is finite, and satisfy the Markov and Martingale properties.

<u>Increment of Brownian motion</u>

Now let's study how Brownian motion behaves for increases.

Finite:

$X(t_1, t_2) = B(t_2) - B(t_1),$

$$X \sim N(0, t_2 - t_1), \ \ t_1 \leq t_2$$

Infinitesimal:

$$dB_t \sim N(0, dt)$$

$$Cov(dB_t, dB_{t'}) = \begin{cases} 0, & t \neq t' \\ dt, & t = t' \end{cases}$$

$$B(T) = B(0) + \int_0^T dB_t$$

<u>Brownian motion with drift</u>

Having defined the standard Brownian motion, I will use it now to create a return generating process with a different drift and volatility. The standard Brownian is based on random standard variables.

Remember that both returns and log-normal can be approximated by a Normal.

$$r_t = \log\left(\frac{S_t}{S_{t-1}}\right) = \mu_0 + \sigma_0 z_t \sim N(\mu_0, \sigma_0^2)$$

Following this equation returns at time T with respect to time 0 (now T represents time, not :

$$\log\left(\frac{S_T}{S_0}\right) = \log\left(\frac{S_T}{S_{T-1}}\frac{S_{T-1}}{S_{T-2}} \dots \frac{S_1}{S_0}\right) = T\mu_0 + \sigma_0\sqrt{T}z_t \sim N(\mu_0, \sigma_0^2)$$

Note that the definition of these returns is for a Δt=1. To achieve our continuous process we need to make $\Delta t \to 0$.

$$\lim_{\Delta t \to 0} [\log\left(\frac{S_T}{S_0}\right)] = \lim_{\Delta t \to 0} [\sum_{t=1}^{T/\Delta t}(\mu_0 \Delta t) + \sum_{t=1}^{T/\Delta t}(\sigma_0\sqrt{\Delta t}z_t)] = \mu_0 T + \sigma \int dB_T \sim N(\mu_0 T, \sigma_0^2 T)$$

To make the last equality clearer: the left hand of the equation becomes $\mu_0 T$ when you develop the summation $\frac{T}{\Delta t} = n\ (number\ of\ steps)$; and the right hand side is a bit more tricky, but is just like applying our case 3 in which we derived the standard Brownian motion.

Finally we can rewrite this last equation for a non-constant drift and volatility, which depend on time.

$$\log\left(\frac{S_T}{S_0}\right) = \int_{t_1}^{t_2} \mu(t)T + \int_{t_1}^{t_2} \sigma(t)dB_T$$

Ito's lemma

An Ito process is a stochastic process of the form:

$dX_t = adt + bdB_t$(B_t is the Brownian motion)

Now assume we have a function of X, F(X(t),t). How would you compute d(F(X))? The traditional calculus would use the chain rule and say:

$dF(t,X) = \frac{\partial F}{\partial t}dt + \frac{\partial F}{\partial dX}dX$

But this is not true, the traditional calculus rules can't be equally applied to stochastic processes don't. It doesn't mean that the traditional calculus is wrong; it means that there exists a generalized form for compute differentials and traditional calculus is one of its cases.

X is nowhere differentiable (due to the Brownian motion) and therefore the usual chain rule doesn't hold, but we can extend it for stochastic variables. I'll derive what is known as Ito's lemma, a key component in Ito Calculus, used to determine the derivative of a time-dependent function of a stochastic process.

X is not differentiable but F(X) is(why?).This would be the standard limit for computing F'(x)

$$F'(x) = \lim_{\Delta x \to 0}\left(\frac{F(x+\Delta x) - F(x)}{\Delta x}\right)$$

I will take the Taylor approximation of $F(X(t),t)$.Remember the definition for a Taylor expansion of 1 variable f(x)

$$\begin{aligned} f(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 \\ &\quad + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \frac{f''''(x_0)}{4!}(x - x_0)^4 + \cdots \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n. \end{aligned}$$

Note that you can pass $f(x_0)$ to the left side thus having $f(x) - f(x_0)$. If we take infinitesimal increases (for which Taylor series are good approximations) now this $f(x) - f(x_0)$ becomes df. Therefore on the right hand-side we would have $(x - x_0)$ that would become dx.

Applying this intuition for a function of two variables I will expand dF to its second order expansion:

$$dF = \left(\frac{\partial F}{\partial t}dt + \frac{\partial F}{\partial X}dX + \frac{1}{2}\frac{\partial^2 F}{\partial X^2}(dX)^2 + \frac{\partial^2 F}{\partial t \partial X}dtdX + \frac{1}{2}\frac{\partial^2 F}{\partial t^2}(dt)^2 + O((dt)^3, (dX)^3, \dots)\right)$$

As a heuristic for this prove, we will keep dF in order of dt, so we can interpret that higher order terms than dt will vanish. All terms from order 3 will vanish and regarding the second order terms we have to take the following into account its expectations:

$$dt^2 \to 0$$

$$dBdt \to 0$$

$$(dB_t)^2 \to dt$$

$$(dX)^2 = a^2dt^2 + 2abdtdB + b^2dB \to b^2dB \to b^2dt$$

$$dtdX = dt(adt + bdB) = adt^2 + bdtdB \to 0$$

$$dt^2 \to 0$$

Therefore all the terms from the $(dX)^2$ will vanish:

$$dF = \left(\frac{\partial F}{\partial t}dt + \frac{\partial F}{\partial X}[\,dX] + \frac{1}{2}\frac{\partial^2 F}{\partial X^2}[dX^2]\right) = \left(\frac{\partial F}{\partial t}dt + \frac{\partial F}{\partial X}[\,dX] + \frac{1}{2}\frac{\partial^2 F}{\partial X^2}[b^2dt]\right)$$

$$\boxed{dF \;= \left[\frac{\partial F}{\partial t} + \frac{b^2}{2}\frac{\partial^2 F}{\partial X^2}\right]dt + \left[\frac{\partial F}{\partial X}\right]dX}$$

Using the definition of $dX_t = \; adt \; + \; b\, dB_t$

$$\boxed{dF = \left[\frac{\partial F}{\partial t} + \frac{b^2}{2}\frac{\partial^2 F}{\partial X^2} + a\frac{\partial F}{\partial X}\right]dt + \left[b\frac{\partial F}{\partial X}\right]dB}$$

Both marked equations stand for Ito's lemma and it is very common to play with both equations. An important observation is that dF, as well as dX, is also an Ito process (when we differentiate a function of an Ito process we get another Ito process).

This proof is not rigorous at all since the trick of it is when we expand the Taylor series we are thinking like if the higher order terms became non-stochastic, and using the heuristic we are able to substitute the remaining terms for its convergence values.

Ito processes examples

Now I will review some of the most know Ito processes to model the behavior of markets and will give a brief explanation and intuition of their utility.

-Brownian motion with drift (arithmetic Brownian motion)

$$dS_t = \mu dt + \sigma dB_t$$

$$S_t = S_0 + \mu T + \sigma(B_t - B_0)$$

But allows negative prices

-Geometric Brownian motion with drift (log normal Brownian motion)

$$dS_t = \mu S_t dt + \sigma S_t dB_t$$

$$d(lnS_t) = (\mu - \frac{\sigma^2}{2})\, dt + \sigma dB_t$$

Prices are modeled to follow log normal paths, integrating the previous expression:

$$\ln(S_t) - \ln(S_0) = \left(\mu - \frac{\sigma^2}{2}\right)T + \sigma d(B_T - B_0)$$

$$S_t = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)T + \sigma d(B_T - B_0)}$$

It excludes negative prices, but drift and variance go to zero as S approaches zeros, and does not allow for bankruptcy or credit defaults (S=0).

-Ornstein-Uhlenbeck process

$$dS_t = \lambda(\overline{S} - S_t)dt + \sigma dB_t$$

It is a mean-reversion process similar to the $AR(1)$. It has no bounds on the values it can take but as further it is from the mean, more "force" it will receive to get return to it. Can be generalized to have a slowly varying mean.

-Cox-Ingersoll-Ross process

$$dr_t = \lambda(\bar{r} - r_t)dt + \sigma\sqrt{r_t}dB_t$$

Another process with mean-reversion mechanics used for interest rates and the term structure (the relationship between short and long interest rates).

Black-Scholes PDE equation

After having understood the Brownian Motion and Ito's Lemma we are ready to derive and understand the famous and the 1997 Nobel Prize winner Black-Scholes equation, demonstrated by Fischer Black and Myron Scholes in 1973. It is a partial differential equation for computing the price of a European option under certain assumptions:

1. Riskless rate: A constant risk-free rate.

2. The stock price follows a geometric Brownian motion with constant drift and volatility.

3. The stock doesn't pay dividend.

4. No arbitrage opportunity (profit without risk).

5. Ability to borrow and lend any amount of cash at the riskless rate. (This is no extra-fees for borrowing)

6. Ability to buy and sell any amount of stocks and options.

7. No transaction costs.

There are different ways in which we could derive it. I will develop 2, the first one is simpler but has a slight error in its derivation (the result is correct), and the second is fully correct.

First proof

Let $V\ (t, S)$ be an option (in fact it could any a derivative that has the same input variables) depending on a stock S, where $ds = (\mu S)dt + (\sigma S)dB$

Applying Ito's Lemma:

$$dV = \left(\frac{\partial V}{\partial t} + \frac{(\sigma S)^2}{2}\frac{\partial^2 V}{\partial S^2}\right)dt + \left(\frac{\partial V}{\partial S}\right)dS$$

Combine the stock and the option in a portfolio π=V-ΔS, where Δ is the number of stocks we hold. We are allowed to modify the weights of the portfolio at anytime.

$$d\pi = dV - \Delta dS = \left(\frac{\partial V}{\partial t} + \frac{(\sigma S)^2}{2}\frac{\partial^2 V}{\partial S^2}\right)dt + \left(\frac{\partial V}{\partial S} - \Delta\right)dS$$

Notice that the only stochastic term is dS. We can make it vanish if $\Delta = \frac{\partial V}{\partial S}$

We have eliminated any risk/randomness, and following the risk-free rate assumption the portfolio will grow at the risk-free rate.

$$d\pi = (r\pi)dt = r(V - \Delta S)dt = (rV - rS\frac{\partial V}{\partial S})dt$$

Equaling the dt coefficients of the two equations below

$$\frac{\partial V}{\partial t} + \frac{(\sigma S)^2}{2}\frac{\partial^2 V}{\partial S^2} = rV - rS\frac{\partial V}{\partial S}$$

$$\boxed{\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{(\sigma S)^2}{2}\frac{\partial^2 V}{\partial S^2} - rV = 0}$$

This is the famous Black-Scholes PDE used to model options prices.

As I said before, this derivation is not fully correct. We defined Δ as the number of stocks, and in one of the steps we had the portfolio π and computed its differential

$$d\pi = dV - \Delta dS$$

This differential would be correct if Δ was a constant, but it is not. We set $\Delta = \frac{\partial V}{\partial S}$ which follows a stochastic behavior that can be differentiated with Ito's calculus.

$\mathrm{d}(\Delta S) = \Delta \mathrm{dS} + \mathrm{Sd}\Delta + \mathrm{d}\Delta\mathrm{dS}$ instead of ΔdS

However this mistake does not carry any error in our calculations and the final result is correct.

<u>Second proof</u>

This second derivation is a bit longer and trickier than the first one but is more rigorous.

Assumptions:

- Unlimited credit to invest or borrow.

-There is a constant and single rate r.

- Stocks can be traded in fractional quantities.

- No dividends.

- No transaction costs.

- No trading delays or market impact.

This part is just to introduce a self-financing condition:

Consider a self-financing portfolio X with initial value zero that holds stocks(S) and bonds (M: this bonds can be viewed as cash receiving free interests) in quantities q and C

$$X_0 = q_0 S_0 + C_0 M_0 = 0$$

Since it is self-financing the initial value of our portfolio is zero. We can rebalance the portfolio at any time t (at the end of a period) borrowing cash to buy more stocks or the opposite. Also from Black-Scholes assumptions remember that there are no arbitrage opportunities and therefore rebalancing the portfolio changing cash for stocks or vice versa won't change the overall value of the portfolio, which is 0.

$$X_t^{post} - X_t^{pre} = S_t\left(q_{post} - q_{pre}\right) + M_t\left(C_{post} - C_{pre}\right) = 0$$

$$(S_t - S_{t-1})\left(q_{post} - q_{pre}\right) + (M_t - M_{t-1})\left(C_{post} - C_{pre}\right) + S_{t-1}\left(q_{post} - q_{pre}\right) + M_{t-1}\left(C_{post} - C_{pre}\right) = 0$$

This last equation is just defined for convenience. If we generalize to continuous time we have this self-financing condition that will be useful later:

$$dSdq + dMdC + Sdq + MdC = 0$$

Now after having derived this condition, assume another portfolio π of three assets: the stock(S) in quantity q, a single derivative(V), and cash M in quantity C. Is like our previous rebalanced portfolio plus V. The last condition also holds for this new portfolio because we could also derive it from it: in a certain time t where prices can't change because we don't move in time, we rebalance only between cash and the stock, letting the quantity for the derivative V constant, which will make it vanish and lead to the same condition.

$$\pi = V + qS + CM$$

The portfolio's change in value between two periods in which there is no rebalancing is

$$\pi_t - \pi_{t-1} = (V_t - V_{t-1}) + q_{t-1}(S_t - S_{t-1}) + C_{t-1}(M_t - M_{t-1})$$

In continuous time

$$d\pi = dV + d(qS + CM) =$$

$$= dV + (qdS + CdM) + dSdq + dMdC + Sdq + MdC =$$

Similar to when I showed that our first proof was not correct, I have used Ito's calculus for deriving the product of two stochastic variables $d(X_tY_t) = Y_tdX_t + X_tdY_t + d[X_t, Y_t]$. Later we will see how to derive this result. I have ordered the terms in a way that fit the self-financing condition that we defined before, so we can delete these terms.

Also notice that since the bonds are cash growing at the risk-free rate r, $dM = rMdt$

$$= dV + (qdS + CdM) = dV + qdS + rCMdt =$$

Finally, from the definition for the portfolio π we can substitute for CM

$$= dV + qdS + r(\pi - V - qS)dt$$

As we did in the first proof, we want to look for a quantity q that eliminates the randomness. Recall that from the Black-Scholes assumptions dV follows the geometric Brownian motion and we can apply Ito's to develop it in terms of $dt$and dS

$$d\pi = \left(\frac{\partial V}{\partial t} + \frac{(\sigma S)^2}{2}\frac{\partial^2 V}{\partial S^2}\right)dt + \left(\frac{\partial V}{\partial S} + q\right)dS + r(\pi - V - qS)dt$$

$$q = -\Delta = -\frac{\partial V}{\partial S}$$

Unlike we did before, now we are not assuming that q is constant.

From the definition of a self-financing portfolio π=0 and to avoid arbitrage $d\pi = 0$. Therefore:

$$d\pi = \left(\frac{\partial V}{\partial t} + \frac{(\sigma S)^2}{2}\frac{\partial^2 V}{\partial S^2} - rV + rS\frac{\partial V}{\partial S}\right)dt = 0$$

$$\boxed{\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{(\sigma S)^2}{2}\frac{\partial^2 V}{\partial S^2} - rV = 0}$$

This election for q is known as dynamic hedging.

<u>Black-Scholes for different assets</u>

In the research literature is very common to start with the original Black-Scholes PDE and stretch some of the assumptions to get even more general results. For example, the assumption that the stock is not paying dividends can be break and get a new solution for a theoretical continuous dividend yield. The portfolio would be no longer self-financing and we would need to take into account the new cash flows:

$$d\pi = \big(rV - \Delta S(r - D)\big)dt$$

$$\frac{\partial V}{\partial t} + \frac{(\sigma S)^2}{2}\frac{\partial^2 V}{\partial S^2} + (r - D)S\frac{\partial V}{\partial S} - rV = 0$$

Also the same PDE can be adapted for different types of derivatives such as:

Currency options (that receive a foreign interest rate)

$$\frac{\partial V}{\partial t} + \frac{(\sigma S)^2}{2}\frac{\partial^2 V}{\partial S^2} + (r - r^*)S\frac{\partial V}{\partial S} - rV = 0$$

Commodity options (that have a cost q of carry the commodity)

$$\frac{\partial V}{\partial t} + \frac{(\sigma S)^2}{2}\frac{\partial^2 V}{\partial S^2} + (r + q - D)S\frac{\partial V}{\partial S} - rV = 0$$

<u>Summary</u>

Ito process:

$$dX_t = adt + bdB_t$$

Ito formula:

$$dF = \left[\frac{\partial F}{\partial t} + \frac{b^2}{2}\frac{\partial^2 F}{\partial X^2}\right]dt + \left[\frac{\partial F}{\partial X}\right]dX$$

$$= \left[\frac{\partial F}{\partial t} + \frac{b^2}{2}\frac{\partial^2 F}{\partial X^2} + a\frac{\partial F}{\partial X}\right]dt + \left[b\frac{\partial F}{\partial X}\right]dB$$

Stock price:

$$dS_t = \mu S_t dt + \sigma S_t dB_t$$

$$d(lnS_t) = (\mu - \frac{\sigma^2}{2})\, dt + \sigma dB_t$$

Black-Scholes:

$$\Delta = \frac{\partial V}{\partial S}$$

$$\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{(\sigma S)^2}{2}\frac{\partial^2 V}{\partial S^2} - rV = 0$$

Fixed income modelling

Interest rates are the price of money. When a bank borrows money to an institution or an individual it does it at a certain price, maybe a 2%.

From a macroeconomic perspective, our current system is based on Keynesian economics: the economic activity of a country is measured with certain indicators that create two different curve, the IS curve and the LM curve.

The former is an equilibrium curve for the goods market, where everything produced is consumed by all the agents of the economy. The intuition for this descendent curve is that when the interest rates are high, companies aren't going to borrow money and the production of goods will be low. When the interests are low, the opposite happens.

The latter is another equilibrium curve but for the monetary markets, all the demanded money equals the money offered in the economy. This is ascending curve and the its rationale is that when the production of a country is high, the income level rises and agents of the economy will ask for more money to make transactions and the interest rates will be high. When the production is low interests will be low.

Although at first it may seem contradictory, but it is not, these curves just represent equilibrium levels for different markets. Both curves can be put together in the IS-LM model that represents a simultaneous equilibrium in both markets.

This equilibrium can be modified by the modification of any of the two curves. There are two types of policies to control this equilibrium: the fiscal policy set by the government changes the IS curve (with for example changes in the spending policies or taxes); and monetary policy controlled by the central bank to change the LM curve (such as modifying the discount rate or purchasing government securities)

Both agents act independently but have to coordinate their actions to try to achieve the desired objectives for the country. So mainly, the interest rates are set by the central bank to let money to the commercial banks and then commercial banks let this money at a higher rate to other economic agents (people, companies, etc.), or offer an interest for borrowing money (for example they give you a 1% for keeping your money deposited so they can lend a part of it to other agents).

As you have seen interest rates are very important in our society and being able to model their behavior has an extremely importance for economic, social and political decisions.

In this part I will explore some of the applications of stochastic and Ito calculus to model fixed income assets. These assets are a debt instrument issued by an "important" institution i.e a government, corporation, or other entity. Basically they need huge amounts of money to do their things and when you buy these bonds you are letting them your money in promise of a future payoff.

Modelling interest rates have a key role in modelling bonds, since they have an inverse relationship. Roughly speaking when interest rates rise up, borrowing money have a higher cost, so the new issued bonds will promise a greater pay in the future than the previous bond. Therefore the previous bond (which is the one you had bought) loses value. You would have to lower your current bond to be able to sell it in the current market, since there are already new bonds paying a greater payoff.

Contrary to what Black-Scholes model assumed, interest rates change over time and also are stochastic. These have a term structure (also known as yield curve) which represents how the interest rates (or yields) of similar quality bonds evolve at different maturities. Finally, remember that interest rates themselves can't be traded; we can trade bonds or interest rate derivatives that directly depend on them.

<u>One-factor model</u>

To begin with our first general model we will try to model bonds in a way such that:

1-There is no arbitrage.

2-There is a fix number of stochastic factors.

We will derive a no arbitrage model where the prices only depend on the short rate (the instantaneous spot interest rate), described by an Ito process

$$dy_t = adt + bdB$$

$$V = V(t, T, y_t)$$

All bond prices are assumed to depend on a single stochastic variable, the short rate mentioned before(y_t). Using two zero coupon bonds of different maturities we can create a risk-free portfolio.

Since each bond depends on a single stochastic variable (y_t), we can apply Ito's lemma

$$dV_i = \left[\frac{\partial V_i}{\partial t} + \frac{b^2}{2}\frac{\partial^2 V_i}{\partial y^2}\right]dt + \frac{\partial V_i}{\partial y}dy$$

π is a portfolio that eliminates risk growing at the risk-free short rate (which is common to assume it equal to the interest rate):

$$\pi = q_1 V_1 + q_2 V_2, \quad where \frac{q_1}{q_2} = -\frac{\frac{\partial V_2}{\partial y}}{\frac{\partial V_1}{\partial y}}$$

$$d\pi = q_1 dV_1 + q_2 dV_2 = y\pi dt$$

$$d\pi = \sum_{i=1,2} q_i \left[\frac{\partial V_i}{\partial t} + \frac{b^2}{2}\frac{\partial^2 V_i}{\partial y^2}\right] dt = y(q_1 V_1 + q_2 V_2)dt$$

For convenience, we will define q_1 and q_2 in a way that will enable to separate V_1 from V_2 in the equation from above:

$$q1 = \frac{1}{\frac{\partial V_1}{\partial y}}$$

$$q2 = -\frac{1}{\frac{\partial V_2}{\partial y}}$$

Equating dt coefficients and separating V_1 terms from V_2:

$$\frac{\frac{\partial V_1}{\partial t} + \frac{b^2}{2}\frac{\partial^2 V_1}{\partial y^2} - yV_1}{\frac{\delta V_1}{\delta y}} = \frac{\frac{\partial V_2}{\partial t} + \frac{b^2}{2}\frac{\partial^2 V_2}{\partial y^2} - yV_2}{\frac{\delta V_2}{\delta y}} = f(t, y)$$

At first you may think that this equality just changed things of place but if you look closely you can get a very useful detail from it: V1 and V_2 are bonds at different maturities (functions of T_1 and T_2 respectively), but how can be possible that the equality above holds if both right and left side depend on its own bond maturity? The fact is that they don't depend. To be able to keep this expression true each side of the equation has been able to cancel the dependence on maturity. Therefore this expression depends only on t and y; and could be applied to any pair of bonds (of any maturity).

$$\frac{\partial V_i}{\partial t} + \frac{b^2}{2}\frac{\partial^2 V_i}{\partial y^2} - yV_1 - f(t, y)\frac{\partial V_i}{\partial y} = 0$$

Rearranging a bit our equation we arrive to a very similar PDE to the BS, but the difference is that the term y (short interest rate) is not constant. From the zero coupon bound definition, the pay at any maturity time T is 1

$$V_i(T, y) = 1$$

The function $f(t,y)$ is unspecified but it can be interpreted as the risk premium for the bond V. As we saw before we can Ito's Lemma for the differential of the bond

$$dy_t = adt + bdB$$

$$dV = \left[\frac{\partial V_i}{\partial t} + \frac{b^2}{2}\frac{\partial^2 V_i}{\partial y^2}\right]dt + \frac{\partial V_i}{\partial y}dy = \left[\frac{\partial V_i}{\partial t} + \frac{b^2}{2}\frac{\partial^2 V_i}{\partial y^2} + a\frac{\partial V_i}{\partial y}\right]dt + \left[b\ \frac{\partial V_i}{\partial y}\right]dB$$

And for the expression that we got before for any pair of bonds we can substitute for $\frac{\partial V_i}{\partial t} + \frac{b^2}{2}\frac{\partial^2 V_i}{\partial y^2}$

$$dV = \left[(V_1 + f(t,y)\frac{\partial V_i}{\partial y})\ + a\frac{\partial V_i}{\partial y}\right]dt + \left[b\ \frac{\partial V_i}{\partial y}\right]dB$$

The coefficient of dB stands for the volatility and dt coefficient is the drift. So, as with the Sharpe ratio we can compute excess return per unit of risk, that also follows an Ito process.

$$\frac{dV - yVdt^*}{b\frac{\partial V}{\partial y}} = \left[\frac{a+f}{b}\right]dt + dB$$

*$yVdt$ is the risk free evolution of the bond.

The dt coefficient represents the market price of risk η

$$\eta = \frac{a+f}{b}$$

It is identical for all bonds and represents the deterministic amount of extra return per unit of randomness. In practice, it is unobservable and we need to either fit parameters form the market observed prices or to make assumptions that lead to tractable solutions.

<u>Modelling the short rate</u>

The only assumption we made before is that the short rate was following an Ito process, but a and b could take any value. Therefore, it is also very important to correctly model the short rate.

Similarly to stocks, there are lots of models for its behavior and I will show some of them.

-<u>Ho & Lee</u>

$$dy = \psi(t)dt + \sigma dB$$

$$\psi(t) = -\frac{\partial^2}{\partial t^2}\ln V_{mkt}(t_0,t) + \sigma^2(t-t_0)$$

Dynamic drift and constant volatility and assumes no arbitrage.
The function $V_{mkt}(t_0, t)$ represents a function of the observed market prices.

-<u>Vasicek</u>

$$dy = \alpha(\bar{y} - y)dt + \sigma dB$$

Modelling interest rates as completely random is might not be a good option, since they don't vary that much over time. A single-factor mean reversion is one way to prevent this diffusion.

-<u>Hull & White</u>

$$dy = [\alpha(\bar{y} - y) + \psi(t)]dt + \sigma dB$$

Combines the two previous models and allows to model both the yield curve and the volatility.

-<u>Cox-Ingersoll-Ross</u>

$$dy = \alpha(\bar{y} - y)dt + \sigma\sqrt{y}dB$$

This last model also modified the volatility making it depend on interest rates.

<u>Black-Scholes PDE</u>

The Black Scholes PDE has a strong relationship with another well-known PDE, the diffusion equation.

$$\frac{\partial p}{\partial t} - \frac{1}{2}\frac{\partial^2 p}{\partial z^2} = 0$$

One of its solutions is the Gaussian distribution

$$p_0(z, t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{z^2}{2t}}$$

This special solution can be used to obtain a general solution.

If $p(z, t = 0) = f(z)$ thegeneral solution is given by

$$p(z, t) = \int p_0(z - z', t) f(z') dz' = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(z-z')^2}{2t}} f(z') dz'$$

The solutions also can be used to price derivatives with any terminal payoff, as an expected value of the derivative:

$$V(S,t) = \int p(S_t, T; S, t) V(S_t, T) dS_t$$

$V(S,t)$: value at time t of a derivative of the stock S

$p(S_t, T; S, t)$:probability at time t of a stock S to reach S_t at time T

$V(S_t, T)$: value at time T of a derivative of the stock S (with its time T value)

<u>Transforming Black-Scholes into the heat equation</u>

A solution to the diffusion equation will give a BS solution via substitution of variables. Terminal values of BS correspond to initial values of the diffusion equation.

A special solution:

$$U(x,\tau) = \frac{1}{\sqrt{2\pi\sigma^2\tau}} e^{-\frac{x^2}{(2\sigma^2\tau)}}$$

From this solution we can obtain a general solution.

<u>Backward equation</u>

A Gaussian random variable $X \backsim N(\mu, \sigma^2)$ has probability density.

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{(2\sigma^2)}}$$

Generalizing for a random walk that begins in a different point and time than the origin:

$$p(x_T, T; x_0, t_0) = \frac{1}{\sqrt{2\pi\sigma^2(T-t_0)}} e^{-\frac{[(x_T-x_0)-\mu(T-t_0)]^2}{2\sigma^2(T-t_0)}}$$

This function satisfies the known backward equation:

$$\frac{\partial p}{\partial t_0} + \frac{\sigma^2}{2}\frac{\partial^2 p}{\partial x_0^2} + \mu\frac{\partial p}{\partial x_0} = 0$$

In the case of pure Brownian motion where μ=0 and σ=1, its probability density obeys the diffusion equation, a PDE that has lots of solutions. A special one that we already have seen is the Gaussian, used to obtain general solutions when given initial conditions.

Survival probabilities

<u>Pure Brownian motion</u>

Assuming pure Brownian motion, what is the probability to get from point A to B without ever hitting C?

I will introduce the method of images: first compute the unrestricted probability of going from A to B. Then subtract the probability of going from A to B hitting point C. To compute this last probability we will compute the probability of going from A* to B without restrictions, where A* is the image point of A with respect to C. For example imagine that A=10, B=12, C=9.

The probability of going from A to B and hit C is the same probability than going from A*=9-(10-9)=8 to B without any restrictions. At first it might seem difficult to get but if you give it a thought in fact it has a lot of sense. From a combinatorics view it means the same: the probability of point A to hit the barrier is the same than A*. At this point paths starting from A and A* are at the same point and we don't need to care anymore about both paths, since from then they do the same trajectory that will end up hitting B.

From the pure Brownian motion we can define the probability of hitting z^* from point z_0 in t time as $p_0(z-z_0,t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(z-z_0)^2}{2t}}$

The probability of going from z to z_0 and hit $z*$ can be defined as $p_0(z-[2z^*-z_0],t)$

Therefore the survival probability density can be defined as

$$p_s(z,t) = p_0(z-z_0,t) - p_0(z-[2z^*-z_0],t) = \frac{1}{\sqrt{2\pi t}}\left(e^{-\frac{(z-z_0)^2}{2t}} - e^{-\frac{(z-[2z^*-z_0])^2}{2t}}\right)$$

Which obeys the boundary condition $p_s(z^*,t) = 0$-> The probability of surviving if you start in z^* is 0. So, generalizing a bit more, for $t > 0$:

$$p_s(z,t) = \begin{cases} \frac{1}{\sqrt{2\pi t}}\left(e^{-\frac{(z-z_0)^2}{2t}} - e^{-\frac{(z-[2z^*-z_0])^2}{2t}}\right) & z > z^* \\ 0 & z < z^* \end{cases}$$

<u>Brownian motion with drift</u>

When we include drift in this problem the symmetry is broken. The new solution is computed as before but now the second term of p_s has a term constant C that could be derived from the boundary conditions $p_s(z^*, t) = 0$: $p_s(z, t) = \ p_0(z - z_0, t) - Cp_0(z - [2z^* - z_0], t)$

This problem is used to compute non-default probabilities of companies, and for addressing similar questions such as how to structure the capital to minimize the default probability.

<u>Solving BS PDE</u>

Let's return to the probability density function of a stock price the follows geometric Brownian motion

$$dS = \mu S dt + \sigma S dB$$

Notice that it satisfies the backward equation, where now drift and volatility are defined by μS and σS. So $p(S_t, T; S, t)$ satisfies

$$\frac{\partial p}{\partial t} + \frac{(\sigma S)^2}{2}\frac{\partial^2 p}{\partial S^2} + \mu S\frac{\partial p}{\partial S} = 0$$

The future expected value of a function of S_t:

$$E[f(S_t)] =$$

So for a vanilla call option of strike K expiring at time T

$$V(S, t) = \frac{e^{-r(T-t)}}{} \int_{x'=lnK}^{\infty} e \ \ \left(e^{x'} - K\right) dx' \ = S\phi(d_+) - Ke^{-r(T-t)}\phi(d_-)$$

Where

$$d_\pm = \frac{\ln\left(\frac{S}{Ke^{-r(T-t)}}\right)}{\sigma\sqrt{T-t}} \pm \frac{1}{2}\sigma\sqrt{T-t}$$

$$\phi(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} e^{-\frac{z^2}{2}}\, dz$$

<u>Monte Carlo simulation</u>

Another way to compute the price of an option is with Monte Carlo simulations. Basically simulate the behavior of the stock and the terminal value of the call, and compute the expected present value over thousands of paths.

The Greeks

These are some partial derivatives of the solution that

$$Delta \ \Delta = \frac{\partial V}{\partial S} = \begin{cases} \phi(d_+) \\ \phi(-d_+) \end{cases}$$

$$Gamma \ \Gamma = \frac{\partial^2 V}{\partial S^2} = \frac{\phi'(d_+)}{\sigma S\sqrt{T-t}}$$

$$Vega \ \nu = \frac{\partial V}{\partial \sigma} = \phi'(d_+)S\sqrt{T-t}$$

Ito in higher dimensions

For combining multiple Ito processes, the formula can be generalized

$$dX_i = a_i(t, X_1, X_2, \dots)dt + b_i(t, X_1, X_2, \dots)dB_i$$

$$dF = \frac{\partial F}{\partial t}dt + \sum \frac{\partial F}{\partial X_i}dX_i + \frac{1}{2}\sum p_{ij}b_i b_j \frac{\partial^2 F}{\partial X_i \partial X_j}dt$$

As an heuristic to complete the formula

$$(dX_i)(dX_j) \to p_{ij}b_i b_j dt$$

$$(dX_i)^2 \to b_i^2 dt$$

For the multiplication of two geometric Brownian motions:

And for the division:

Linear Algebra for asset pricing

In this part I will explore some of the applications of Linear Algebra in Finance.

One-period model

Payoff matrix

I will start explaining the one-period model. In this model we represent the payoff of different securities (these are financial assets that can be traded) in different states (viewed as scenarios) of the world from one time to another.

For example imagine four securities: a bond that is riskless and always pays $1; a stock that pays $3 in the first scenario/state, $2 in the second and $1 in the third; and 2 other calls with different payoffs.

Each state happens has associated different probabilities. State 1 has p=½, state 2 has p=1/6 and state 3 has p=1/3.

State	Probability	Bond	Stock	Call 1	Call 2
1	1/2	$1	$3	$1.5	$2
2	1/6	$1	$2	$0.5	$1
3	1/3	$1	$1	$0	$0

The payoff of these securities can be represented in a payoff matrix:

$$A = \begin{bmatrix} 1 & 3 & 1.5 & 2 \\ 1 & 2 & 0.5 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

This matrix A acts like a linear transformation $R^n \text{ -> } R^s$, where n is the number of securities and s the number of states. Applying this linear transformation to a vector $x \in R^n$ (that represents your portfolio: how many of each securities do you have), you get the payoff of this portfolio at each state. For example a portfolio of 1 stock would be x=[0, 1,0,0] and applying the transformation Ax, we would get the payoff for each state which in this case is the same as the stock payoff.

Now let's try to find a certain payoff b for a given number of securities. We have $Ax = b$, where x is our unknown.

If A is invertible (non-singular), there one solution $x = A^{-1}b$. This means that there are no redundant securities and that $n = s$.

If there are more states than securities $(n < s)$ in general there is no solution unless $b \in Image(A)$. Anyways we could obtain the minimum norm solution for x using the pseudoinverse of A: $(A^T A)^{-1} A^T$

If $n > s$ there could be a solution or not. If there is, there are infinite ones which can be found adding any element of $Nul(A)$. Since in this case A is not square we can use its pseudoinverse to make any necessary computation (instead of deleting all the redundant securities).

<u>Market completeness</u>

In a complete market there exists a portfolio that can generate any payoff. Algebraically, it has the following implications:

- $Im(A) =$ full space of payoffs

- $rank(A) = s$, # of independent states of the world

- A is surjective

If $n = s$ and A is invertible, there is a unique solution for any b. If n>s it might be the case that we could drop the redundant securities and still be able to invert A, being therefore in a complete market. Otherwise it is incomplete, since there are some payoffs that cannot be generated with available assets. Equivalently:

- $Im(A) =$ a subspace of the space generated by the payoffs

- $rank(A) < s$

This notation allows creating portfolios that match a desired payoff. It is very useful to compute a hedge strategies (when it is possible), and to correct for deviations from a theoretical solution.

<u>Redundant securities</u>

We may find redundant securities in any type of market. A security is redundant if we can create its same payoff as a linear combination of other securities. Since securities are written in the columns, it is the same as having linear dependent columns. Algebraically:

- $Nul(A)$ is non-empty (different than 0)

- There can exist arbitrage portfolios, which we will see later

When there are redundant securities, for a given portfolio x, $Ax = b$ the payoff is unique, but for a given payoff b there can be different portfolios x that satisfy it. As there are infinite solutions for the $Nul(A)$, we can apply linearity for finding different portfolios with the same payoff:

$$z \in Ker(A)$$

$$A(x + cz) = Ax = b \; for \; all \; c$$

<u>Market value of a portfolio</u>

Given the market value of n securities, we can organize them in a S matrix (n x 1). The market value of a portfolio of these assets can be computed as:

$$MV = \sum_{i=1}^{n} S_i x_i = (S_1 \, S_2 \, \dots S_n)\begin{pmatrix} x_1 \\ \cdots \\ x_n \end{pmatrix} = S^t x$$

<u>Arbitrage</u>

In this model we can find two types of arbitrage:

Type 1 arbitrage: a portfolio of MV 0 that has at least a payoff in one of the states>0 and all the other payoffs>=0. This is that you may earn some money, but in the worst scenario you stay the same.

Type 2 arbitrage: a portfolio of MV>0 that gives a payoff of 0 at every state. This means that you could made a portfolio that will not pay nothing but sell it at non-zero price.

No-arbitrage pricing

Under the no-arbitrage assumption, there are new concepts that can be introduced.

- Law of one price:

This law states that a security with a certain payoff will have the same price that any other security with the same payoff. This law doesn't allow arbitrage.

Consequently, asset prices are a linear combination of the payoffs at each state and we can find a global price for something. With arbitrage we couldn't look for that convergence in prices and we can price any portfolio without finding contradictions.

- Relative pricing vs absolute pricing:

Relative pricing: Given some assets and their price we can establish a relationship for a new dependent asset, or at least a bound for its price. If it is independent the price has to be given externally. In other words, we compare how something is valued in one places to offer a price according to them.

Absolute pricing: aims to fix prices in terms external variables. For example try to value a stock analyzing the balance sheet, earnings report, etc.

The concepts given in this section are focused on relative rather than absolute pricing.

- Arrow-Debreu securities

These are portfolios such that the combination of its assets gives a 1 unit payoff in a single state and 0 in the others.

An elementary AD security can be replicated if there is a portfolio such that:

$$Ax = \begin{pmatrix} 0 \\ 0 \\ \dots \\ 1 \\ \dots \\ 0 \end{pmatrix}$$

The prices of each elementary AD security are the state prices.

$$\psi = \begin{pmatrix} \psi_1 \\ \dots \\ \psi_S \end{pmatrix}$$

The state prices can be interpreted as the price of each monetary unit at each state. For example ψ_1 corresponds to the prices of the fist AD security, which is the portfolio that gives 1 unit of payoff at state 1 and 0 at the others. Therefore the price of a monetary unit at state 1 corresponds to ψ_1.

Remember that there could be more than one portfolio replicating the same AD security since we can add elements belonging to Nul(A).

State prices must be positive since by definition, the payoffs are positive. Under no arbitrage:

- You can take AD securities as basis assets

- Payoff matrix for AD securities A=I.

- Price for any positive payoff is $\psi^T\, b > 0$

- The price for any asset of the portfolio can be obtained from the state prices $S_1 = \psi^T\, a_1, etc$

$$S^T = \psi^T\, A$$

In complete markets is easy to compute the state prices, but for incomplete markets there could be infinite solutions for state prices:

- If there is at least one solution that has all the state prices greater than 0, $\psi\ > 0$, there is no arbitrage.

- If every solution has $\psi\ \leq 0$ then there are arbitrage opportunities.

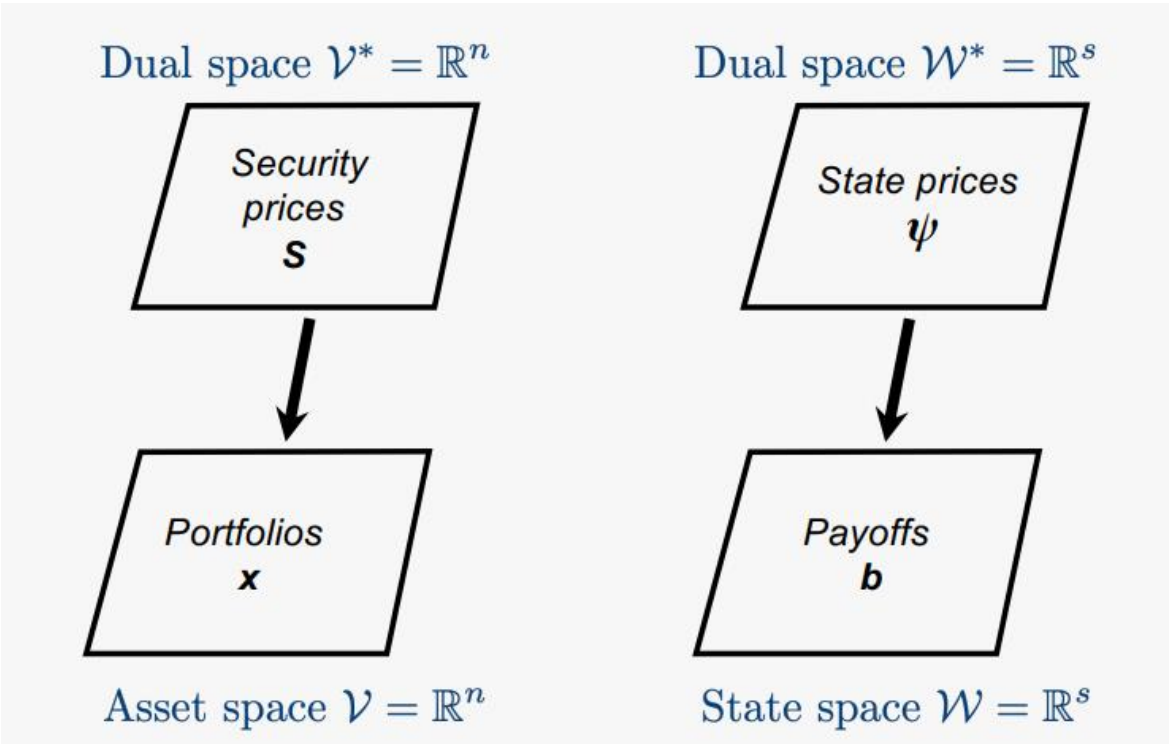
So in summary, given a payoff matrix A, prices S and a target of payoffs b:

Finding all $\psi\ > 0$ such that $S^T = \psi^T\, A$ can result in

1- No solutions -> arbitrage
2- 1 solution -> complete market
3- Multiple solutions ->incomplete market
    a. We price assets using all solutions:
        Since it is an incomplete market, if we get one solution it means that there is a redundant asset.
        Otherwise, due to incompleteness you can find a market in which different state prices are consistent with no arbitrage.

- Dual Spaces

We can see these concepts of security prices S, state prices ψ, portfolios x and payoffs b combined in what is known as dual spaces, using the payoff matrix A.

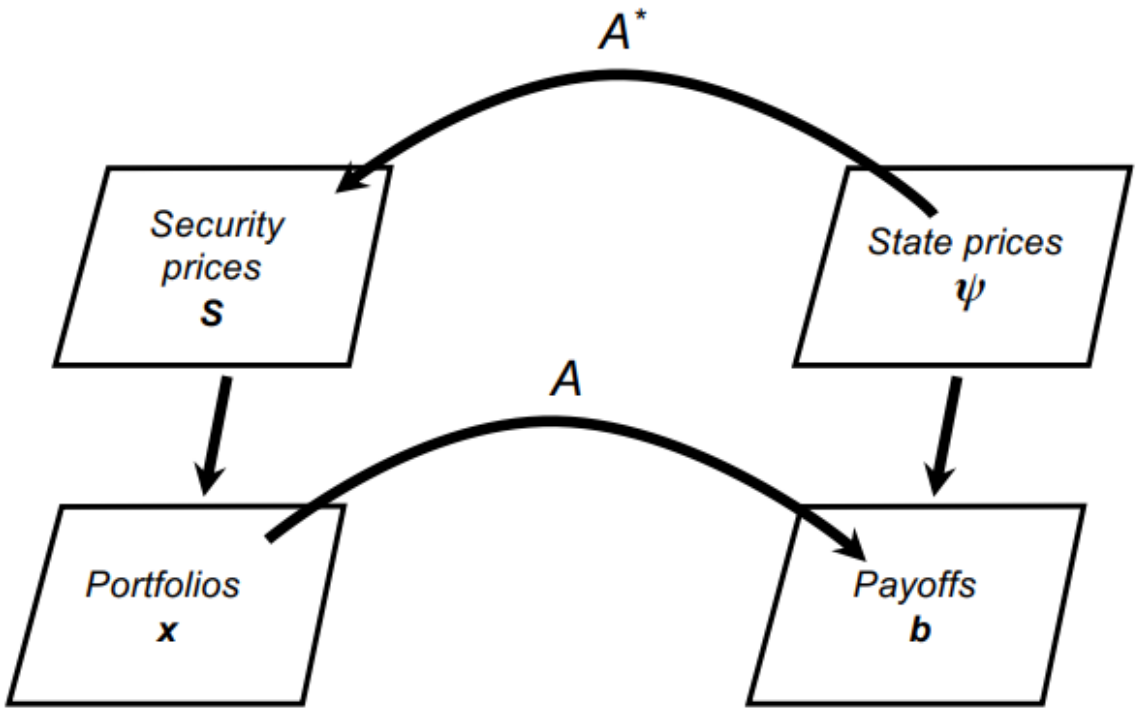


There are two spaces.

The asset space (R^n), in which we have the security prices S and a portfolio x. We can compute the price of the portfolio using $S[x] = S^T x$.

The state space (R^S), in which we have the state prices () and a payoff b. We can compute the state price of the payoff using $\psi[b] = \psi^T b$.

We can relate these spaces between them using the payoff matrix A:



Given a portfolio x we can apply the linear transformation $A(n{-}{>}\ s)$ to obtain the payoffs b.

Given the state prices ψ we we can apply the linear transformation $A^T(s{-}{>}\ n)$ to obtain the security prices S.

Following from the no arbitrage assumption the market value of a portfolio must be the same as the state price of its payoff

$$S[x] = \psi[b]$$

- Fundamental theorem of asset pricing

If there exists a strictly positive state-price vector consistent with the security-price vector, there is no arbitrage:

Absence of type-1 arbitrage (proof):

Absence of type-2 arbitrage(proof):

If there is no arbitrage, there exists a strictly positive state-price vector consistent with the security-price vector

This proof follows form Farka's lemma which leads to one of these options, but not both:

1-

2-

<u>Optimization</u>

<u>Defining a portfolio of assets</u>

When we think about portfolio optimization, we should find an optimal trade-off between risk and return. Generally we have a budget constrain, and we have to choose the weights (w) to distribute between different assets. Provided that all assets have mean and volatility, this election of weights will create a portfolio with certain return and risk.

Portfolio return:

The return of a portfolio will be the sum of the returns of each asset, but weighted by the weights.

$$\mu_p = \mu^T w$$

Portfolio variance:

The variance is not that intuitively to get. For example, for two assets remember that the variance is: $Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$.

So in the simplest case covariances=0, and we only had to sum its variances (which is almost never true). Therefore, if we generalize for any number of assets you will notice that the variance can be expressed as:

$$\sigma_p^2 = w^T C w = s \sum w_i^2 \sigma_i^2 + 2 \sum_{i<j} w_i w_j \sigma_i \sigma_j p_{ij}$$

Where C is the covariance matrix between all the assets, which is symmetric and positive-definite (because we know that the variances, which are the diagonal terms, are positive). The positive definite property is only guaranteed if we exclude: risk-free assets, these would be independent to any asset and had a variance of 0; and linearly dependent assets that would made the matrix less than full rank.

Budget constraints:

Our objective is to maximize the return and minimize the risk under some constraints. Typically the weights have to sum up 1, but it is not mandatory. This constraint is represented as $\sum_i w_i = 1$

Solving the problem

Maximize the mean and minimize the variance admits a lot of different interpretations. Specifically, I am going to focus on minimizing the variance of the portfolio

$$minimize\ f(w) = \frac{1}{2} w^T C w$$

$$such\ that\ \iota^T w = 1$$

ι^T is just a vector of 1's that allow us to express the summation as a matrix multiplication. This type of problems are well-known to be solved using Lagrange multipliers and if we need to generalize, the KKT condition

$$L(w, \lambda) = \frac{1}{2} w^T C w + \lambda(1 - \iota^T w)$$

$$\frac{\partial L}{\partial w_i} = \sum_j^n C_{ij} w_j - \lambda \iota_i = 0 \rightarrow Cw - \lambda \iota = 0$$

$$w = \lambda C^{-1} \iota$$

Now solve for the Lagrange multiplier

$$\frac{\partial L}{\partial \lambda} = \iota^T w - 1 = 0 \rightarrow \lambda(\iota^T C^{-1} \iota) = 1 \rightarrow \lambda = \frac{1}{(\iota^T C^{-1} \iota)}$$

$$w_{min} = \lambda C^{-1} \iota = \frac{C^{-1} \iota}{(\iota^T C^{-1} \iota)}$$

$$\sigma^2 = w^T C w = (\iota^T {C^{-1}}^T \lambda)\ C\ (\lambda C^{-1} \iota) = \lambda^2 \left(\iota^T {C^{-1}}^T \iota\right)^* = \frac{\lambda^2}{\lambda} = \lambda$$

*It is easy to proof that $\iota^T {C^{-1}}^T \iota = \iota^T C^{-1} \iota = \frac{1}{\lambda}$ since ι is a vector of 1's and in both equalities represent a summation of all the elements of the matrix C^{-1}

Return constraints

Now I am going to minimize the variance for a given portfolio mean, which is we are going to add a new constraint on the mean: $\mu_p = \mu^T w$

Solving the problem

$$L(w, \lambda, m) = \frac{1}{2} w^T C w + \lambda(1 - \iota^T w) + m(\mu_p - \mu^T w)$$

$$\frac{\partial L}{\partial w_i} = \sum_j^n C_{ij} w_j - \lambda \iota_i - m\mu_i = 0 \rightarrow \; Cw - \lambda\iota - m\mu = 0$$

Since C is invertible (because is positive definite)

$$w = C^{-1}(\lambda\iota + m\mu)$$

Now, let's compute the other two Lagrangian derivatives and substitute with the result for w:

$$\frac{\partial L}{\partial \lambda} = (1 - \iota^T w) = 0$$

$$\iota^T w = \lambda(\iota^T C^{-1}\iota) + m(\mu^T C^{-1}\iota) = 1$$

$$\frac{\partial L}{\partial m} = \left(\mu_p - \mu^T w\right) = 0$$

$$\mu_p^T w = \lambda(\mu^T C^{-1}\iota) + m(\mu^T C^{-1}\mu) = \mu_p$$

We can group this two equations in a matrix form:

$$\begin{pmatrix} 1 \\ \mu_p \end{pmatrix} = M \begin{pmatrix} \lambda \\ m \end{pmatrix}$$

$$M = \begin{pmatrix} a & b \\ b & c \end{pmatrix} = \begin{pmatrix} \iota^T C^{-1}\iota & \mu^T C^{-1}\iota \\ \mu^T C^{-1}\iota & \mu^T C^{-1}\mu \end{pmatrix}$$

And then solve for the multipliers:

$$\begin{pmatrix} \lambda \\ m \end{pmatrix} = M^{-1} \begin{pmatrix} 1 \\ \mu_p \end{pmatrix} = \frac{1}{ac - b^2} \begin{pmatrix} c - b\mu_p \\ -b + a\mu_p \end{pmatrix}$$

Finally, we can compute the weights

$$w = C^{-1}(\lambda\iota + m\mu)$$

And compute the variance of the minimized portfolio

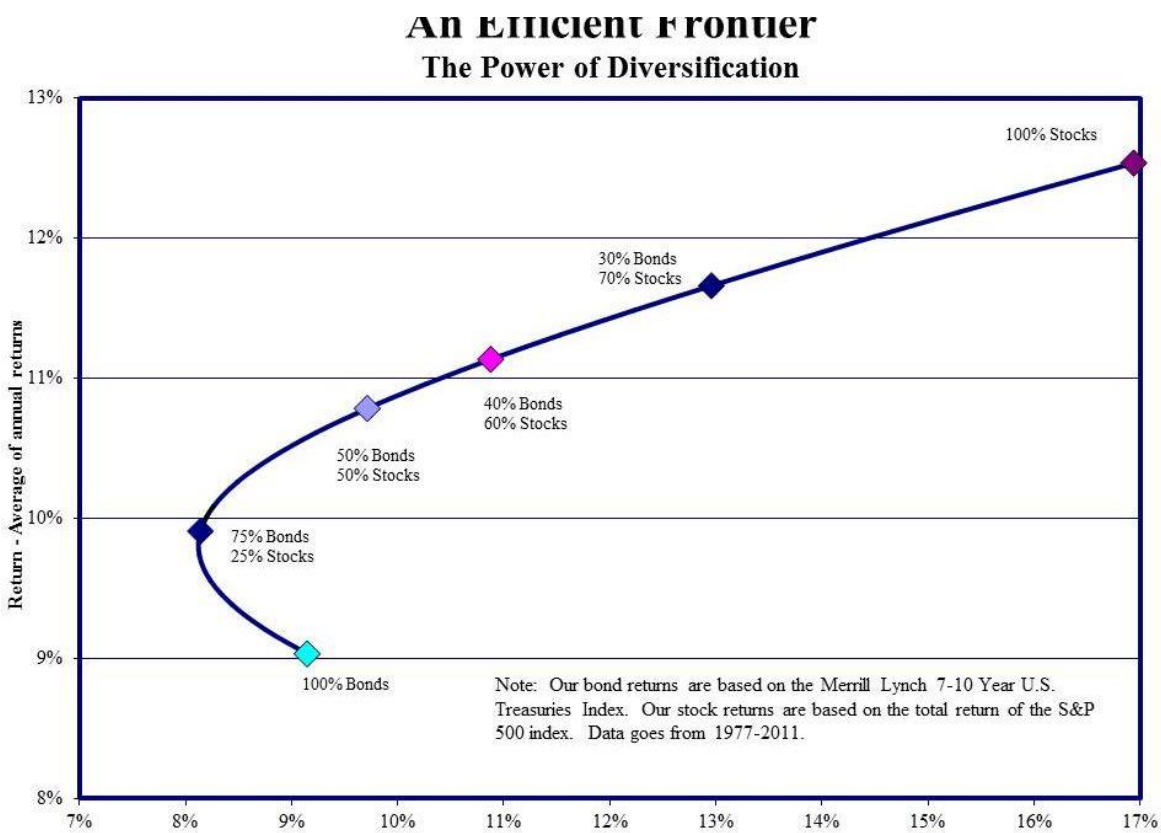
$$\sigma^2 = w^T C w = w^T C C^{-1}(\lambda\iota + m\mu) = w^T(\lambda\iota + m\mu) = (\lambda\iota + m\mu)^T {C^{-1}}^T(\lambda\iota + m\mu) =$$
$$= (\lambda\iota + m\mu)^T({C^{-1}}^T\lambda\iota + {C^{-1}}^T m\mu) = (\lambda\iota^T + m\mu^T)({C^{-1}}^T\lambda\iota + {C^{-1}}^T m\mu) =$$
$$= \lambda\iota^T {C^{-1}}^T\lambda\iota + \lambda\iota^T {C^{-1}}^T m\mu + m\mu^T {C^{-1}}^T\lambda\iota + m\mu^T {C^{-1}}^T m\mu =$$

$$= \lambda^2 a + \lambda m b + m\lambda b + m^2 c =$$

$$= (\lambda \quad m) M \begin{pmatrix} \lambda \\ m \end{pmatrix}$$

$$\sigma^2 = (\lambda \quad m)\begin{pmatrix} 1 \\ \mu_p \end{pmatrix} = \frac{1}{ac - b^2}(a\mu_p^2 - 2b\mu_p + c)$$

<u>Efficient frontier</u>



This function for the volatility vs the return for different weights is known as the efficient frontier. Plotted for different weights you can observe the curvature you usually get for representing mean and variance, which show the importance of diversification

Every combination at the right of the frontier would be achievable since it would be a worse portfolio, but there is no way to reach any point at the left of the frontier since the frontier plots the most optimized portfolios.

Portfolios in the efficient frontier follow the solution we found before for w which is

$$w = C^{-1}(\lambda\iota + m\mu)$$

It is a linear combination of two portfolios, also known as the two characteristic portfolios: the first $C^{-1}\lambda\iota$ is the minimum variance portfolio that we could get, while the second $C^{-1}m\mu$ si the prortfolio of the maximum Sharpe ratio.

This problem is called a single-period optimization problem because is only valid for the time we plot the frontier. In practice this would hold if everything is stationary and the covariance matrix is both accurate and stationary. When things change over time (as it usually happens) results are not fixed anymore, and in practice the real problem lies on estimating the parameters correctly (mean, variance, covariance).

The results can be generalized in two ways. First, we can add more Lagrange multipliers to the equation and solve in the same way we have done until now. Second, there can be inequality constraints: you may say that want all weights to be <0 and <1, so you have no negative weights (short positions) and no weights higher than 1(leveraged portfolio); or maybe you would like to define that the weight for a certain asset should be higher than a value. You can add almost any constraint you like but these are inequalities, therefore to solve them you can use the KKT conditions.

In practice to solve any of these optimization problems you can use any program which will handle the constraints for you.

Finally, note that:

1-It is informative to plot your portfolios vs the frontier to know if a portfolio is near the frontier or there is room for improvement.

2- Is worth to study the how sensitive is the optimized output changing depending on the input, since it is probable that your approximations for the parameters will have some errors. So if changing a bit your parameters gives a completely different portfolio you should take this variability into account.